Velo Working Pte. Ltd.

Fabrication & Production Module

Business Plan & Vendor Reference — Internal Document

Prepared by: Ronnie Yap, Founder
Date: May 2026
Status: Internal Reference — Not for External Distribution

1. Executive Summary

Velo Working's Fabrication & Production module exists to solve one of the most persistent pain points for engineering and industrial founders in Singapore and Malaysia: the inability to access high-quality, affordable hardware production without either paying premium local prices or exposing themselves to the risk of managing overseas vendors directly.

We act as a single-point production partner — absorbing the complexity of vendor selection, overseas sourcing, quality assurance, and logistics — and delivering finished hardware to our clients on specification and on schedule. Our model is built on a curated network of proven manufacturers in China, India, and Singapore, giving clients access to a global production capability through one trusted local interface.

The module covers five production categories: structural metal fabrication, cable harnessing and wiring assembly, control panel engineering, PCB and electronic assembly, and new energy infrastructure hardware. Each category is backed by a dedicated vendor, selected for proven quality, cost efficiency, and reliability.

2. The Problem

Founders running engineering and manufacturing businesses in Singapore face a structural cost and capability problem when it comes to hardware production. The challenge is not a lack of options — it is the cost and complexity of navigating them.

High Local Production Costs

- Local Singapore fabricators and cable assembly houses price at a significant premium due to high overheads, labour costs, and land. For SMEs, this directly erodes project margin and makes competitive quoting difficult.
- Control panel and automation hardware sourced through local distributors carries heavy markups on top of already expensive brand names.

Overseas Sourcing Is Risky Without the Right Network

- Going directly to Chinese or Indian manufacturers requires established relationships, technical fluency, and on-the-ground quality oversight — resources most SME founders do not have.
- Without a trusted intermediary, founders face inconsistent quality, communication gaps, delayed shipments, and no recourse when specifications are missed.

- Managing three to five vendors for a single project — one for metal, one for cabling, one for panels — creates coordination overhead that most lean teams cannot absorb.

The Capability Gap

- Most hardware founders are strong on engineering but stretched on procurement, vendor management, and supply chain. They need a partner who speaks their language technically, but who takes the sourcing burden off their plate entirely.
- Smaller companies in particular struggle to access the same supplier quality that larger contractors can, simply because they lack the order volume to build those relationships independently.

3. Our Strategy

Velo Working positions itself as a production management partner, not a manufacturer. This distinction is the core of the business model.

Single Point of Accountability

The client deals only with Velo Working. We scope the requirement, select the appropriate vendor, manage the production schedule, conduct quality assurance, and arrange delivery. The client receives a finished product — not a vendor relationship to manage.

Curated Vendor Network — Pre-Qualified, Not Ad Hoc

Our vendor network is not a marketplace. Each partner has been selected based on proven capability, quality track record, and cost efficiency. The four anchor vendors — Huahui Shanghai, Actenn, Grantech, and iEminence Engineering — cover the full spectrum of our production categories. When a job comes in, we know exactly who to call.

Cost Arbitrage Passed to the Client

By sourcing through manufacturers in China and India where appropriate, and leveraging Singapore-based partners for precision and compliance-sensitive work, we offer clients a cost structure that local-only sourcing cannot match. This is not about cutting corners — it is about placing the right job with the right vendor at the right price point.

Lean Internal Structure = Competitive Pricing

Velo Working does not carry the overhead of a full manufacturing operation — no factory floor, no large permanent technical headcount. Our internal team focuses purely on project management, procurement, and quality assurance. This keeps our cost base low and our pricing sharp.

4. What We Can Produce

The following five production categories represent Velo Working's current scope of supply. Each category is backed by one or more anchor vendors.

A. Structural Metal Fabrication

Custom metal structures, enclosures, and assemblies for industrial and commercial applications.

- Steel structures and custom metal assemblies
- Electrical enclosures and control boxes
- Architectural aluminium panels and fluorocarbon metal cladding
- Rail transit components — platform screen doors (full-height and half-height), cable trays, signal cabinets
- Curtain wall systems for commercial and infrastructure projects
- New energy infrastructure enclosures — EV charging cabinets, energy storage unit control cabinets, lithium battery equipment housings

Primary vendor: Huahui Shanghai (China) | Supporting vendor: Grantech (Singapore)

B. Cable Harnessing & Wiring Assembly

End-to-end cable assembly for industrial, automotive, defense, and commercial applications — from single harnesses to batch production runs.

- Industrial wire harnesses (single and multi-branch)
- Automotive-grade cable assemblies with EMI shielding and chemical resistance
- Military and defense-grade cables and harnesses (high-temperature, EMI-protected)
- Custom power cables
- Fiber optic and defense optics cabling
- Complex cable harnessing for manufacturing and production equipment

Primary vendor: Actenn (Singapore) | Supporting vendor: Grantech (Singapore)

C. Control Panel Engineering

Design and build of industrial control panels and automation hardware — from relay logic to full PLC-driven control systems.

- Industrial control panels and remote I/O (RIO) panels
- Relay logic panels
- Stainless steel panels for clean or harsh environments
- Pre-wired and box build control assemblies
- Ex-rated junction boxes for hazardous area applications
- Schneider PLC systems: M580, M340, M241, M251, M221 series
- Magelis HMI systems
- Altivar variable frequency drive systems
- Components: Wago, Weidmuller, Eaton

Primary vendor: iEminence Engineering (India) | Supporting vendor: Actenn (Singapore)

D. PCB & Electronic Assembly

Printed circuit board assembly and electronic box build for industrial, semiconductor, and commercial applications.

- SMT (surface mount technology) PCB assembly
- Through-hole and mixed-technology PCB production
- Box build electronics assembly
- Industrial and defense-grade electronics
- Biometric, medical, and railway sector electronics

Primary vendor: Actenn (Singapore) | Supporting vendor: Grantech (Singapore)

E. New Energy Infrastructure

Enclosures, cabinets, and structural hardware for new energy and sustainability-driven projects.

- EV charging station cabinets and enclosures
- Energy storage unit (ESU) control cabinets
- Lithium battery coating oven equipment housing
- Grid-tied energy infrastructure structural components

Primary vendor: Huahui Shanghai (China)

5. Vendor Network Reference

The table below is the primary vendor reference for production engagements. When a job is confirmed, refer to this table to identify the right vendor to engage first.

Vendor	Location	Core Capabilities	Best Engaged For
Huahui Shanghai	China	Rail transit components, platform screen doors, cable trays, signal cabinets, architectural metal panels (aluminium, fluorocarbon), curtain wall systems, new energy cabinets (EV charging, energy storage)	Large structural metal works, enclosures, energy infrastructure, rail & transit projects
Actenn	Singapore	Wire harnesses (industrial, automotive, military/defense), custom cable assembly, PCB assembly, power cables, box build & control panels, defense optics & fiber technology	Cable harnessing, defense-grade wiring, PCB assembly, control panel box builds
Grantech	Singapore	SMT PCB assembly, complex cable harnessing for manufacturing, general metal fabrication, complex assemblies and fixtures	Mixed projects requiring PCB + cable + metal under one vendor
iEminence Engineering	India	Control panel engineering (control panels, RIO panels, relay logic panels, SS panels), Schneider PLCs (M580, M340, M221), HMI, Altivar drives, ex-junction boxes, Wago / Weidmuller / Eaton components	Industrial automation panels, PLC systems, drives, control room builds

6. Production Capability Matrix

Quick-reference matrix mapping each product or service category to the vendor(s) who can fulfil it.

Product / Service Category	Huahui	Actenn	Grantech	iEminence
Steel Structures & Custom Assemblies	✓	–	✓	–
Electrical Enclosures & Control Boxes	✓	–	✓	✓
Architectural Metal Panels (Aluminium / FC)	✓	–	–	–
Rail Transit Components & Platform Screen Doors	✓	–	–	–
New Energy Cabinets (EV / Energy Storage)	✓	–	–	–
Curtain Wall Systems	✓	–	–	–
Wire Harnesses (Industrial / Automotive)	–	✓	✓	–
Military / Defense-Grade Cable Assemblies	–	✓	–	–
Custom Power Cables	–	✓	–	–
PCB Assembly (SMT)	–	✓	✓	–
Box Build & Pre-Wired Control Assemblies	–	✓	–	✓
Industrial Control Panels & RIO Panels	–	–	–	✓
Relay Logic & Stainless Steel Panels	–	–	–	✓
Schneider PLC Systems (M580 / M340 / M221)	–	–	–	✓
HMI & Altivar Drive Systems	–	–	–	✓
Ex-Junction Boxes	–	–	–	✓
Wago / Weidmuller / Eaton Components	–	–	–	✓

(✓) = Primary or capable vendor. (–) = Not in scope for this vendor.

7. Target Clients

Velo Working's fabrication module is most relevant to the following buyer profiles:

- System integrators requiring fabricated enclosures, control panels, and cable assemblies for their client projects.
- Industrial automation companies building or upgrading control rooms, PLC systems, and drive infrastructure.
- Engineering and manufacturing SMEs that outsource hardware production rather than investing in in-house capability.

- Offshore and marine contractors requiring rugged cable assemblies, junction boxes, and custom enclosures.
- Rail and infrastructure contractors needing platform screen doors, cable trays, signal enclosures, and architectural cladding.
- Data centre builders requiring control panels, power distribution hardware, and structural enclosures.
- New energy and EV infrastructure developers requiring certified enclosures and energy storage hardware.

8. Delivery Model

Every engagement follows a consistent delivery process, regardless of production category:

Step 1 — Scope & Requirement Capture

Velo Working meets with the client to understand the full technical requirement — drawings, specifications, quantities, and delivery timeline. We do not pass raw client briefs to vendors; we translate and own the specification.

Step 2 — Vendor Selection & Quotation

Based on the capability matrix, the appropriate vendor(s) are engaged for quotation. We consolidate vendor quotes into a single client-facing proposal, with our project management and QA margin built in.

Step 3 — Production Management

Once the order is confirmed, we manage the full production schedule — including material sourcing, production milestones, and factory-level QA checks where required.

Step 4 — Quality Assurance

Prior to shipment or delivery, all items are inspected against the agreed specification. We hold the vendor accountable for non-conformance, not the client.

Step 5 — Delivery & Handover

Hardware is delivered physically to the client's site or warehouse, with full delivery documentation. For clients with ongoing production requirements, we establish a repeat order schedule to streamline future procurement.

9. Pricing Reference

All pricing is project-based and quote-driven. The ranges below are indicative for planning purposes.

- Standard Fabrication (enclosures, assemblies, steel structures): SGD 2,000 – SGD 20,000 per job
- Cable Harnessing & Assembly (batch): SGD 500 – SGD 8,000 per batch. Volume discounts for recurring orders.
- Control Panel Engineering (industrial panels, PLC builds): SGD 5,000 – SGD 40,000 per panel system
- PCB Assembly (batch): SGD 1,000 – SGD 15,000 depending on complexity and volume
- New Energy Infrastructure (enclosures, cabinets): SGD 10,000 – SGD 50,000 per unit or installation

- Large-Scale Production Contract (Year 2 target): SGD 300,000+ (multi-category, multi-vendor engagement)
- Complex Integrated Contract (Year 3 target): SGD 500,000+ (full hardware supply for major projects)

Service bundling is available — clients requiring multiple production categories (e.g., metal fabrication + cable harnessing + control panels for a single facility) benefit from consolidated project management and a single delivery schedule.

10. Goals & Milestones

Year 1 — Build & Validate

- Secure first fabrication or cable harnessing production order.
- Establish live working relationships with all four anchor vendors.
- Complete at least two client deliveries across different production categories.
- Build first case study for portfolio and marketing use.
- Target: SGD 80,000 total revenue (fabrication module contribution).

Year 2 — Scale

- Land a SGD 300,000 production contract — large-scale or multi-category engagement.
- Introduce quality documentation and SOP for each vendor engagement type.
- Begin Malaysia market outreach leveraging existing industry relationships.
- Onboard procurement and QA staff to support volume growth.

Year 3 — Optimise

- Secure contracts above SGD 500,000 in total fabrication module revenue.
- Optimise vendor terms and pricing through volume commitments.
- Establish Australia market presence (exploratory).
- Improve gross margin through better vendor negotiation and repeat-order efficiencies.

Prepared by: Ronnie Yap, Founder — Velo Working Pte. Ltd.
Address: 540 Sims Ave, Sims Avenue Centre, #03-05, Singapore 387603
Date: May 2026

Signature: ______________________________